

Math 3680: Week 6

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Contents

1	Piecewise functions with Cases	1
2	Matrix multiplication	1
3	Custom commands	2
3.1	Mdframed definition improvements	2
3.2	Figure Improvements	3
3.3	A math command	3

1 Piecewise functions with Cases

Here is a piecewise function with three cases ranging on the domain $([-5, 5])$.

$$f(x) = \begin{cases} x + 2 & \text{if } -5 \leq x < -1 \\ x^2 & \text{if } -1 \leq x \leq 2 \\ 3 - x & \text{if } 2 < x \leq 5 \end{cases}$$

Here is a table of values to go along with it!

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
$f(x)$	-3	-2	-1	0	1	0	1	4	0	-1	-2

2 Matrix multiplication

I have a million of these in my notes preformatted in obsidian-based $\text{\LaTeX}$.

Here is an expanded version of the cross product formula:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \left\langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \right\rangle$$

This is the basic structure for matrix addition:

$$\begin{aligned}
 & \begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1n} \\ x_{21} & x_{22} & x_{23} & \cdots & \vdots \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ x_{m1} & x_{m2} & x_{m3} & \cdots & x_{mn} \end{bmatrix} \\
 + & \begin{bmatrix} y_{11} & y_{12} & y_{13} & \cdots & y_{1n} \\ y_{21} & y_{22} & y_{23} & \cdots & \vdots \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ y_{m1} & y_{m2} & y_{m3} & \cdots & y_{mn} \end{bmatrix} \\
 = & \begin{bmatrix} x_{11} + y_{11} & x_{12} + y_{12} & x_{13} + y_{13} & \cdots & x_{1n} + y_{1n} \\ x_{21} + y_{21} & x_{22} + y_{22} & x_{23} + y_{23} & \cdots & \vdots \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ x_{m1} + y_{m1} & x_{m2} + y_{m2} & x_{m3} + y_{m3} & \cdots & x_{mn} + y_{mn} \end{bmatrix}
 \end{aligned}$$

And here is a matrix problem from my notes that results in two cases (used for eigen solving).

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} -x_1 + 2x_2 = 0 \\ 2x_1 - 4x_2 = 0 \end{cases}$$

3 Custom commands

This can be super useful to me because I have not actually defined custom commands in my work repository, etc.

3.1 Mdframed definition improvements

Looking at a lot of the repeated code in my workbooks, here are a few I might implement in the future: The `\begin{mdframed}` is used 300+ times in my notes/work repository, An example being the definition of variance in statistics:

Definition of Variance

If you have n samples $x_1, x_2, \dots, x_n$ that have a mean of μ , the *variance* is defined to be:

$$v = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

A new command could be defined as

```
\newenvironment{definitionbox}[1]{\begin{mdframed}[style=important,frametitle={#1}]}{\end{mdframed}}
```

To call this new environment, you would use:

Definition of Mean

If you have n samples $x_1, x_2, \dots, x_n$ that have a mean of μ , the *variance* is defined to be:

$$v = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

It is not too much improvement but It allows for more readability and consistency when updating.

3.2 Figure Improvements

A second case of bulk command use would be helpful is the images that we insert into the book.

Each image is inserted into a figure environment, such as

```
% \begin{figure}[H]
% \centering
% \includegraphics[width=0.5\textwidth]{balloon.png}
% \caption{Air pushes both ways on a balloon.}
% \label{fig:balloon}
% \end{figure}
```

The things that change here are the image file, the image width, the caption, and the label. This could be a four input custom command for figures. Potentially, what I am looking at is:

```
\newcommand{\bookfigure}[4][0.75]{% \begin{figure}[H] \centering
\includegraphics[width=#1\textwidth]{#2} \caption{#3} \label{fig:#4}
\end{figure} %}
```

3.3 A math command

This one is a small one for in line one-line vectors (x, y) generally:

```
\newcommand{\vecb}[1]{\begin{bmatrix} #1 \end{bmatrix}}
```

this can be done here as:

$$a_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$